matter taking shape even if you'd been sitting front-row center. You couldn't have seen where the galaxy clusters and voids were starting to form. Before anybody could have seen anything worth seeing, photons had to travel, unimpeded, across the universe, as carriers of this information.

The spot where each photon began its cross-cosmos journey is where it had smacked into the last electron that would ever stand in its way—the "point of last scatter." As more and more photons escape unsmacked, they create an expanding "surface" of last scatter, some 120,000 years deep. That surface is where all the atoms in the universe were born: an electron joins an atomic nucleus, and a little pulse of energy in the form of a photon soars away into the wild red yonder.

By then, some regions of the universe had already begun to coalesce by the gravitational attraction of their parts. Photons that last scattered off electrons in these regions developed a different, slightly cooler profile than those scattering off the less sociable electrons sitting in the middle of nowhere. Where matter accumulated, the strength of gravity grew, enabling more and more matter to gather. These regions seeded the formation of galaxy superclusters while other regions were left relatively empty.

When you map the cosmic microwave background in detail, you find that it's not completely smooth. It's got spots that are slightly hotter and slightly cooler than average. By studying these temperature variations in the CMB—that is to say, by studying patterns in the surface of last scatter—we can infer what the structure and content of the matter was in the early universe. To figure out how galaxies and clusters and superclusters arose, we use our best probe, the CMB—a potent time capsule that empowers astrophysicists to reconstruct cosmic history in reverse. Studying its patterns is like performing some sort of cosmic phrenology, as we analyze the skull bumps of the infant universe.

When constrained by other observations of the contemporary and distant universe, the CMB enables you to decode all sorts of fundamental cosmic properties. Compare the distribution of sizes and temperatures of the warm and cool areas and you can infer how strong the force of gravity was at the time and how quickly matter accumulated, allowing you to then deduce how much ordinary matter, dark matter, and *dark energy* there is in the universe. From here, it's then straightforward to tell whether or not the universe will expand forever.

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